

## Nursing Dx:

- Imbalanced Nutrition: less than body requirements
- Activity intolerance
- Risk for impaired liver function

## Nursing Goals/Planning:

- Relief of discomfort, resume ABLs, return to normal liver function w/out complications, adequate intake of food/fluids

## Nursing Interventions:

- Encourage a well balanced diet
- Teach client to avoid alcohol
- Explain importance of rest
- Educate how to prevent transmission of disease
- Educate about complications
- Instruct client to keep F/U appointments
- Teach to take precautions when traveling
- Provide information on vaccines / HBV & HCV should not donate blood

## Drug Therapy:

- Antivirals (Direct-acting)
- Nucleoside/Nucleotide analogs
- Interferon
- Immune Globulin
- Antipruritic/Antiemetics

## Nutritional Therapy:

- Balanced, nutritional diet/adequate calories
- small, frequent meals
- hydration/ fluid and electrolyte balance

## Liver Biopsy: pre and postop care

Hepatocytes become targets for a virus/toxin

- viral hepatitis
- alcohol hepatitis
- drug-induced hepatitis

ETIOLOGY/  
PATHOPHYSIOLOGY

NURSING  
IMPLICATIONS

HEPATITIS

Risk Factors

ASSESSMENT/CLINICAL  
MANIFESTATIONS

DIAGNOSTICS/LABS

- AST/ALT/ GGT/FibroSure
- Alkaline phosphatase
- Bilirubin
- Coagulation studies
- Serum proteins
- Liver biopsy/ultrasound
- Anti-HAV IgM & IgG, HBsAg, Anti-HBs, anti-HCV, HCV RNA, Anti-HDV, anti-HEV
- alcohol level

- Unscreened blood transfusion
- Hemodialysis
- Crowded conditions
- Poor hygiene of food handlers
- Poor sanitation
- eating uncooked/undercooked meat
- drinking unclean water
- healthcare workers/IV drug users
- Drug overdose/excessive alcohol

- Ask about living conditions
- Assess risky sexual behavior/IV drug use
- Ask about meal preparation
- Ask about client's work
- Ask about recent travels
- Ask about medications/alcohol use

Complete physical assessment/vitals

Assess for:

- Acute
- jaundice
  - nausea/vomiting/loss of appetite
  - flu-like symptoms/fatigue
  - pruritus
  - urine/stools/bowel habits
  - aches and pains
  - enlarged liver/spleen
- Chronic
- jaundice/edema
  - palmar erythema/spider angiomas
  - confusion/asterixis
  - bruising/bleeding

## HEPATITIS A

Transmission: Fecal-oral primarily

Causes: ingestion of contaminated food/water, contact with infected stool

Risk Factors: crowded conditions, poor hygiene of food handlers, poor sanitation

Incubation: 15-50 days

Prevention: Good handwashing, HAV vaccine

HAV vaccine: need 2 doses at least 6 months apart.

## HEPATITIS B

Transmission: Blood/body fluids

Causes: unprotected sex with infected individual, infants born to infected mothers, contact with infected blood, substance use/contaminated needle

Risk Factors: Health care workers, intravenous drug abusers, individuals who live with a person who has HBV, individuals who undergo dialysis

Incubation: 45-180 days

Prevention: Good handwashing, HBV vaccine, needle precautions, avoiding unprotected contact with body fluids of infected people, blood donor screening, testing women who are pregnant

## HEPATITIS E

Transmission: Fecal-oral

Causes: ingestion of food or water contaminated with fecal waste, consumption of uncooked/undercooked meat

Risk Factors: same as HAV

Incubation: 15-64 days

Prevention: Good handwashing, avoid uncooked/undercooked meat. No vaccine in the U.S.

## HEPATITIS D

Transmission: Coinfection with HBV, same as HBV

Causes: substance use/contaminated needles, unprotected sex with infected individual

Risk Factors: similar to HBV except that HDV is uncommon in the U.S.

Incubation: 2-26 weeks

Prevention: Good handwashing, HBV vaccine (no vaccine for HDV), needle precautions, avoiding unprotected contact with body fluids of infected people

## HEPATITIS C

Transmission: Blood/body fluids

Causes: unprotected sex with infected individual, substance use/contaminated needle stick, contaminated tattoo equipment, blood products (contaminated), organ transplants

Risk Factors: Health care workers, intravenous drug users, high-risk sexual practices, blood transfusions administered before 1992

Incubation: 14-180 days

Prevention: Good handwashing, blood donor screening, needle precautions, avoid unprotected sex with infected people. No vaccine available.

## COMPLICATIONS

- CHRONIC HEPATITIS
- FULMINANT HEPATITIS
- CIRRHOSIS OF THE LIVER
- LIVER FAILURE
- LIVER CANCER
- HEPATIC ENCEPHALOPATHY